The LIFEREG Procedure

Model Information	
Data Set	WORK.SURVIVAL_ANALYSIS
Dependent Variable	Log(tenure)
Censoring Variable	censor
Censoring Value(s)	1
Number of Observations	300
Noncensored Values	249
Right Censored Values	51
Left Censored Values	0
Interval Censored Values	0
Number of Parameters	2
Name of Distribution	Weibull
Log Likelihood	-290.9609661

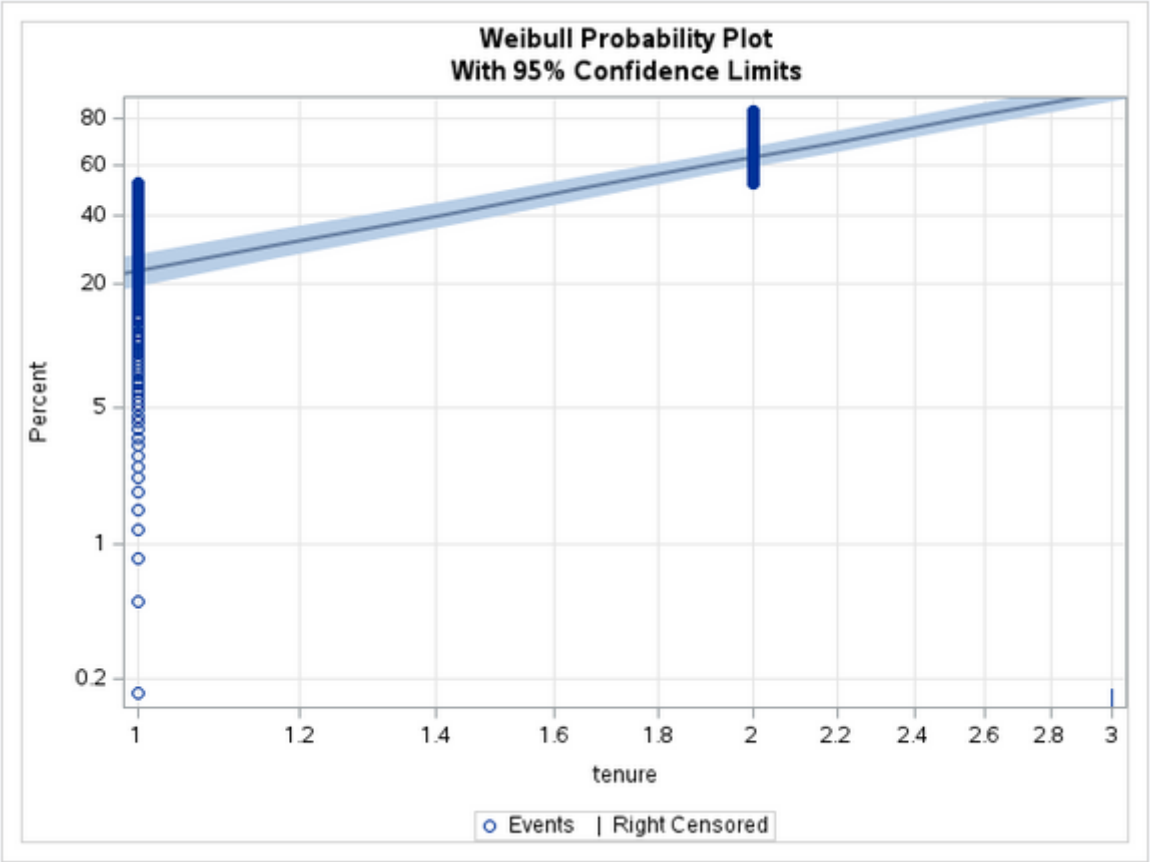
Number of Observations Read	300
Number of Observations Used	300

Fit Statistics	
-2 Log Likelihood	581.922
AIC (smaller is better)	585.922
AICC (smaller is better)	585.962
BIC (smaller is better)	593.329

Fit Statistics (Unlogged Response)	
-2 Log Likelihood	709.461
Weibull AIC (smaller is better)	713.461
Weibull AICC (smaller is better)	713.501
Weibull BIC (smaller is better)	720.869

Algorithm converged.

Analysis of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	0.6855	0.0326	0.6216	0.7493	443.07	<.0001
Scale	1	0.5099	0.0257	0.4618	0.5629		
Weibull Scale	1	1.9847	0.0646	1.8620	2.1155		
Weibull Shape	1	1.9613	0.0991	1.7765	2.1654		



The LIFEREG Procedure

Model Information	
Data Set	WORK.SURVIVAL_ANALYSIS
Dependent Variable	Log(tenure)
Censoring Variable	censor
Censoring Value(s)	1
Number of Observations	300
Noncensored Values	249
Right Censored Values	51
Left Censored Values	0
Interval Censored Values	0
Number of Parameters	1
Name of Distribution	Exponential
Log Likelihood	-355.8160319

Number of Observations Read	300
Number of Observations Used	300

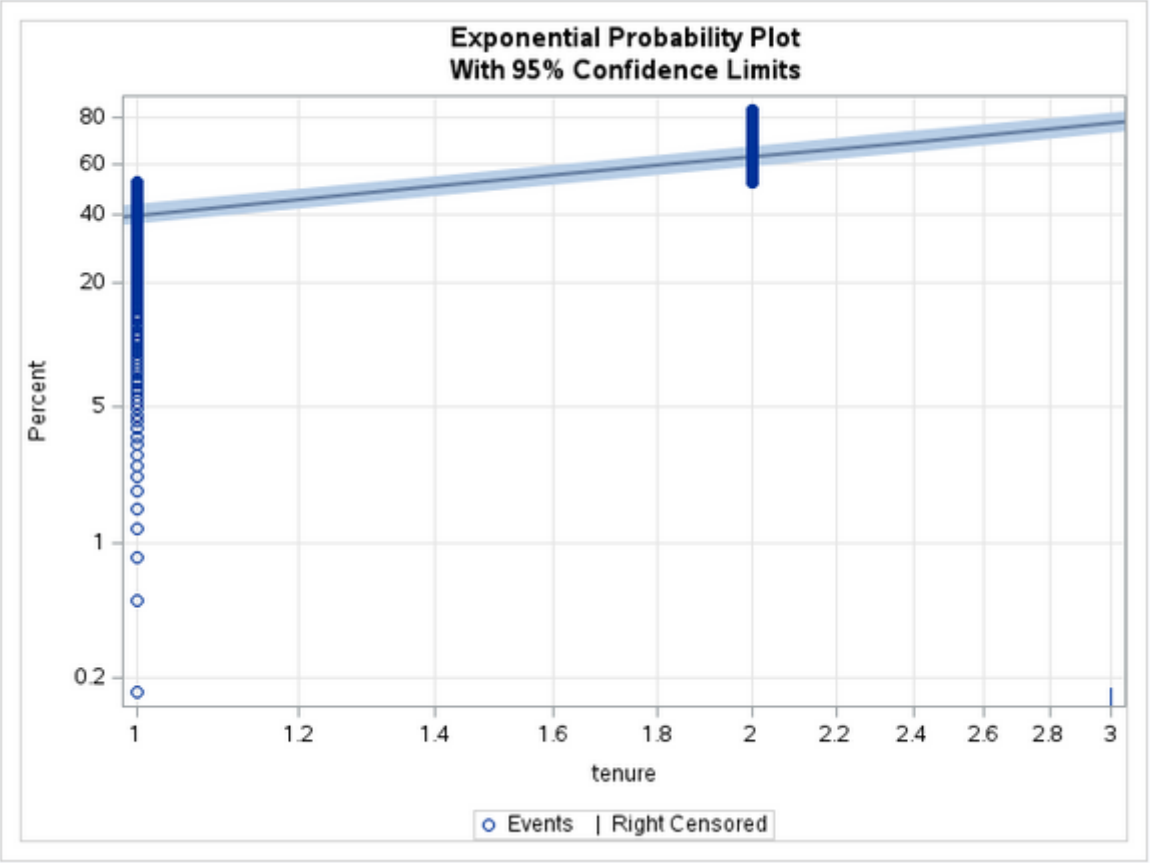
Fit Statistics	
-2 Log Likelihood	711.632
AIC (smaller is better)	713.632
AICC (smaller is better)	713.645
BIC (smaller is better)	717.336

Fit Statistics (Unlogged Response)	
-2 Log Likelihood	839.171
Exponential AIC (smaller is better)	841.171
Exponential AICC (smaller is better)	841.185
Exponential BIC (smaller is better)	844.875

Algorithm converged.

Analysis of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	0.6851	0.0634	0.5609	0.8093	116.87	<.0001
Scale	0	1.0000	0.0000	1.0000	1.0000		
Weibull Scale	1	1.9839	0.1257	1.7522	2.2463		
Weibull Shape	0	1.0000	0.0000	1.0000	1.0000		

Lagrange Multiplier Statistics		
Parameter	Chi-Square	Pr > ChiSq
Scale	.	.



The LIFEREG Procedure

Model Information	
Data Set	WORK.SURVIVAL_ANALYSIS
Dependent Variable	Log(tenure)
Censoring Variable	censor
Censoring Value(s)	1
Number of Observations	300
Noncensored Values	249
Right Censored Values	51
Left Censored Values	0
Interval Censored Values	0
Number of Parameters	9
Name of Distribution	Weibull
Log Likelihood	-274.1497417

Number of Observations Read	300
Number of Observations Used	300

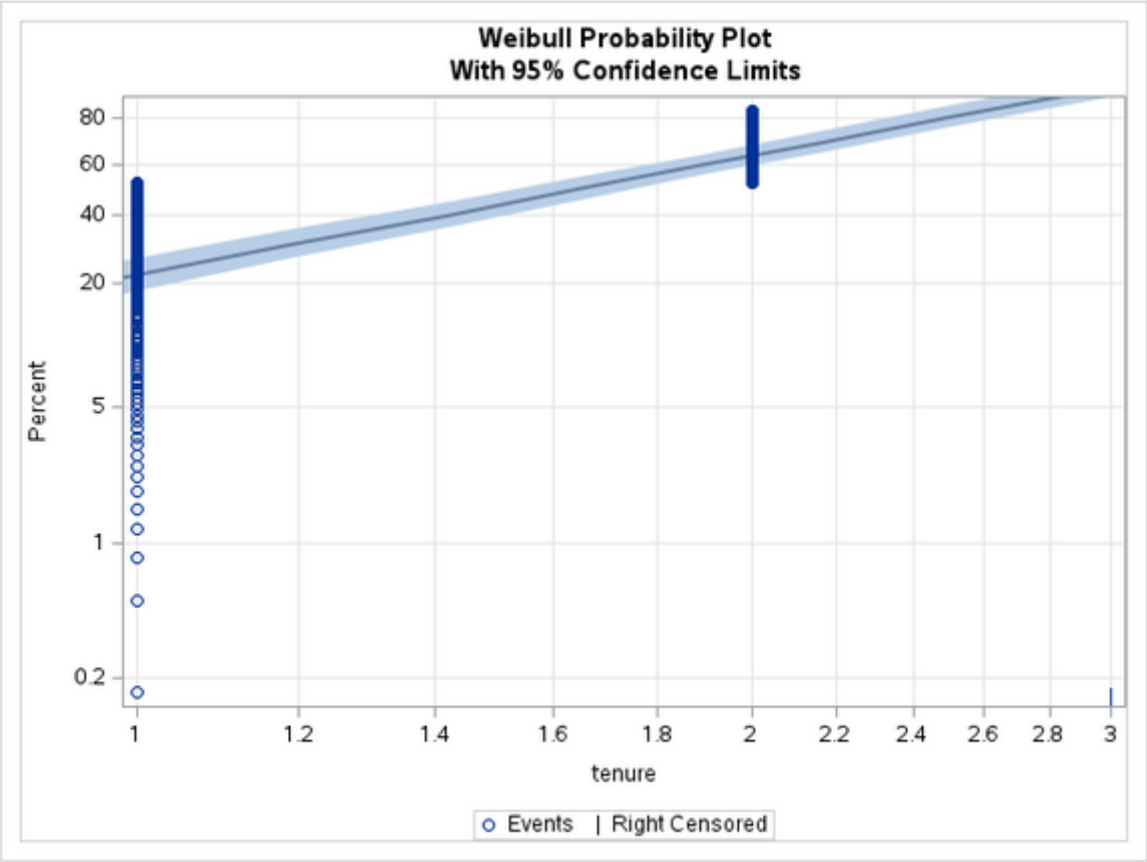
Fit Statistics	
-2 Log Likelihood	548.299
AIC (smaller is better)	566.299
AICC (smaller is better)	566.920
BIC (smaller is better)	599.634

Fit Statistics (Unlogged Response)	
-2 Log Likelihood	675.839
Weibull AIC (smaller is better)	693.839
Weibull AICC (smaller is better)	694.459
Weibull BIC (smaller is better)	727.173

Algorithm converged.

Type III Analysis of Effects			
Effect	DF	Wald Chi-Square	Pr > ChiSq
Risk_appetite	1	13.7616	0.0002
FUND_PERFORMANCE	1	0.9309	0.3346
Inv_potential	1	1.4379	0.2305
Inv_involvement	1	0.5752	0.4482
AUM	1	7.3933	0.0065
Complex_prod	1	0.9679	0.3252
Complaints	1	5.1270	0.0236

Analysis of Maximum Likelihood Parameter Estimates							
Parameter	DF	Estimate	Standard Error	95% Confidence Limits		Chi-Square	Pr > ChiSq
Intercept	1	1.2370	0.1962	0.8525	1.6215	39.77	<.0001
Risk_appetite	1	-0.1582	0.0427	-0.2419	-0.0746	13.76	0.0002
FUND_PERFORMANCE	1	0.0439	0.0455	-0.0452	0.1329	0.93	0.3346
Inv_potential	1	0.0728	0.0607	-0.0462	0.1918	1.44	0.2305
Inv_involvement	1	0.0380	0.0501	-0.0602	0.1361	0.58	0.4482
AUM	1	-0.0927	0.0341	-0.1595	-0.0259	7.39	0.0065
Complex_prod	1	-0.0743	0.0755	-0.2223	0.0737	0.97	0.3252
Complaints	1	-0.2784	0.1230	-0.5194	-0.0374	5.13	0.0236
Scale	1	0.4852	0.0243	0.4399	0.5353		
Weibull Shape	1	2.0608	0.1032	1.8682	2.2733		



The PHREG Procedure

Model Information	
Data Set	WORK.SURVIVAL_ANALYSIS
Dependent Variable	tenure
Censoring Variable	censor
Censoring Value(s)	1
Ties Handling	BRESLOW

Number of Observations Read	300
Number of Observations Used	300

Summary of the Number of Event and Censored Values			
Total	Event	Censored	Percent Censored
300	249	51	17.00

Analysis of Effects Eligible for Entry			
Effect	DF	Score Chi-Square	Pr > ChiSq
Risk_appetite	1	1.9940	0.1579
FUND_PERFORMANCE	1	2.0002	0.1573
Inv_potential	1	2.4563	0.1171
Inv_involvement	1	1.7550	0.1852
AUM	1	0.2854	0.5932
Complex_prod	1	1.1598	0.2815
Complaints	1	2.8198	0.0931

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
10.5171	7	0.1611

Step 1. Effect Complaints is entered. The model contains the following effects:

Complaints

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	2704.151	2701.024
AIC	2704.151	2703.024
SBC	2704.151	2706.541

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	3.1275	1	0.0770
Score	2.8198	1	0.0931
Wald	2.7810	1	0.0954

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Complaints	1	0.40827	0.24482	2.7810	0.0954	1.504

Analysis of Effects Eligible for Entry			
Effect	DF	Score Chi-Square	Pr > ChiSq
Risk_appetite	1	2.1510	0.1425
FUND_PERFORMANCE	1	1.1303	0.2877
Inv_potential	1	1.4847	0.2230
Inv_involvement	1	1.0555	0.3043
AUM	1	0.5263	0.4682
Complex_prod	1	0.9641	0.3262

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
7.8287	6	0.2509

Step 2. Effect Risk_appetite is entered. The model contains the following effects:

Risk_appetite Complaints

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	2704.151	2698.871
AIC	2704.151	2702.871
SBC	2704.151	2709.905

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	5.2805	2	0.0713
Score	4.9666	2	0.0835
Wald	4.9244	2	0.0852

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Risk_appetite	1	0.11645	0.07949	2.1465	0.1429	1.124
Complaints	1	0.41802	0.24492	2.9131	0.0879	1.519

Analysis of Effects Eligible for Entry			
Effect	DF	Score Chi-Square	Pr > ChiSq
FUND_PERFORMANCE	1	1.7184	0.1899
Inv_potential	1	3.3254	0.0682
Inv_involvement	1	2.0169	0.1556
AUM	1	0.7132	0.3984
Complex_prod	1	1.5876	0.2077

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
5.7644	5	0.3298

Step 3. Effect Inv_potential is entered. The model contains the following effects:

Risk_appetite Inv_potential Complaints

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	2704.151	2695.468
AIC	2704.151	2701.468
SBC	2704.151	2712.020

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	8.6835	3	0.0338
Score	8.2547	3	0.0410
Wald	8.1974	3	0.0421

Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Risk_appetite	1	0.17066	0.08500	4.0308	0.0447	1.186
Inv_potential	1	-0.15493	0.08508	3.3158	0.0686	0.856
Complaints	1	0.32460	0.24990	1.6872	0.1940	1.383

Step 4. Effect Complaints is removed. The model contains the following effects:

Risk_appetite Inv_potential

Convergence Status
Convergence criterion (GCONV=1E-8) satisfied.

Model Fit Statistics		
Criterion	Without Covariates	With Covariates
-2 LOG L	2704.151	2697.306
AIC	2704.151	2701.306
SBC	2704.151	2708.341

Testing Global Null Hypothesis: BETA=0			
Test	Chi-Square	DF	Pr > ChiSq
Likelihood Ratio	6.8453	2	0.0326
Score	6.6839	2	0.0354
Wald	6.6634	2	0.0357

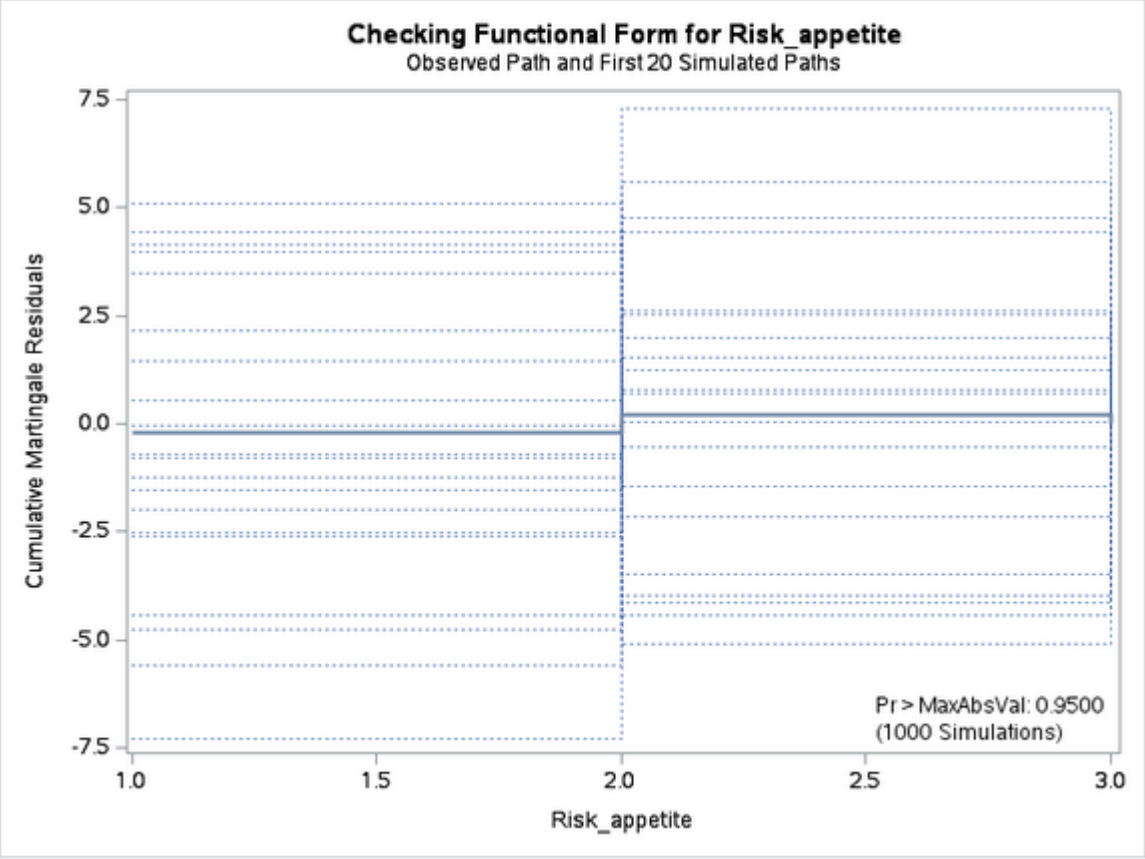
Analysis of Maximum Likelihood Estimates						
Parameter	DF	Parameter Estimate	Standard Error	Chi-Square	Pr > ChiSq	Hazard Ratio
Risk_appetite	1	0.17662	0.08493	4.3249	0.0376	1.193
Inv_potential	1	-0.17984	0.08301	4.6941	0.0303	0.835

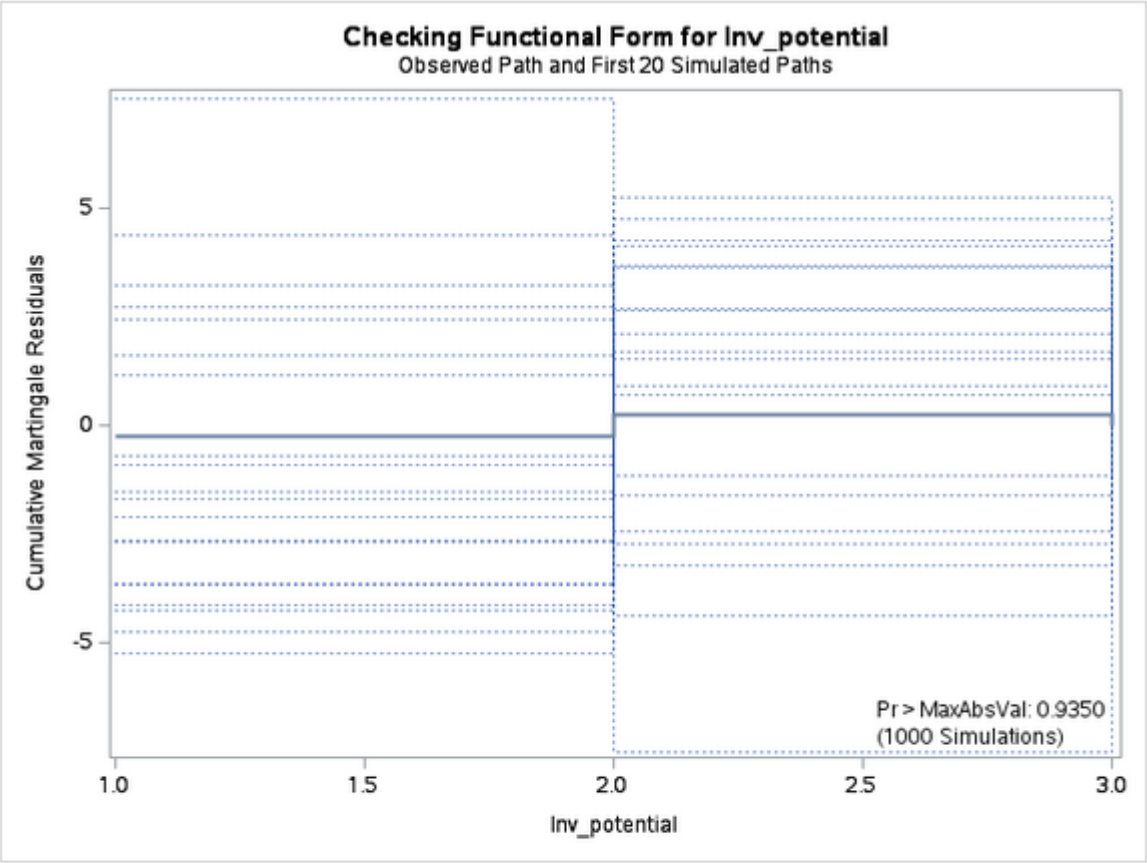
Analysis of Effects Eligible for Entry			
Effect	DF	Score Chi-Square	Pr > ChiSq
FUND_PERFORMANCE	1	0.1662	0.6836
Inv_involvement	1	0.2037	0.6517
AUM	1	1.6390	0.2005
Complex_prod	1	0.2946	0.5873
Complaints	1	1.7011	0.1921

Residual Chi-Square Test		
Chi-Square	DF	Pr > ChiSq
4.1883	5	0.5226

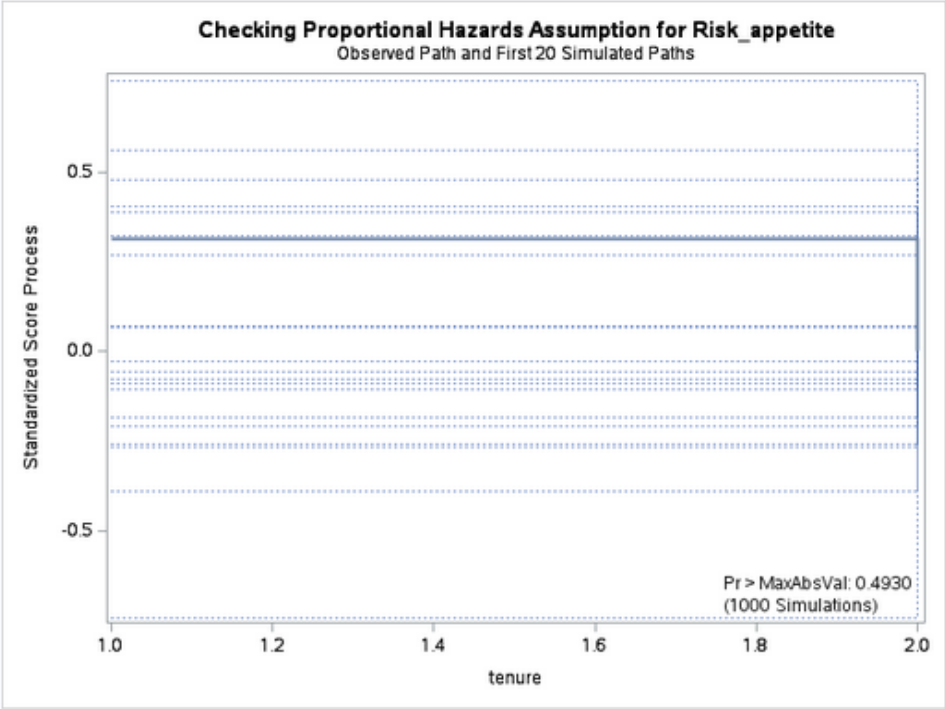
Note: Model building terminates because the effect to be entered is the effect that was removed in the last step.

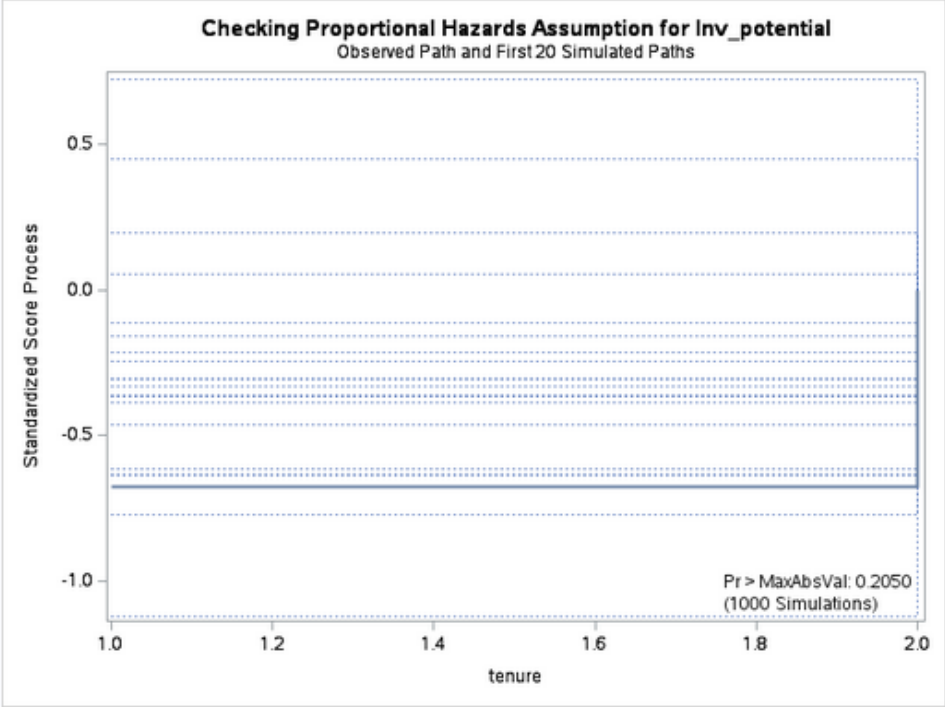
Summary of Stepwise Selection							
	Effect						
Step	Entered	Removed	DF	Number In	Score Chi-Square	Wald Chi-Square	Pr > ChiSq
1	Complaints		1	1	2.8198		0.0931
2	Risk_appetite		1	2	2.1510		0.1425
3	Inv_potential		1	3	3.3254		0.0682
4		Complaints	1	2		1.6872	0.1940



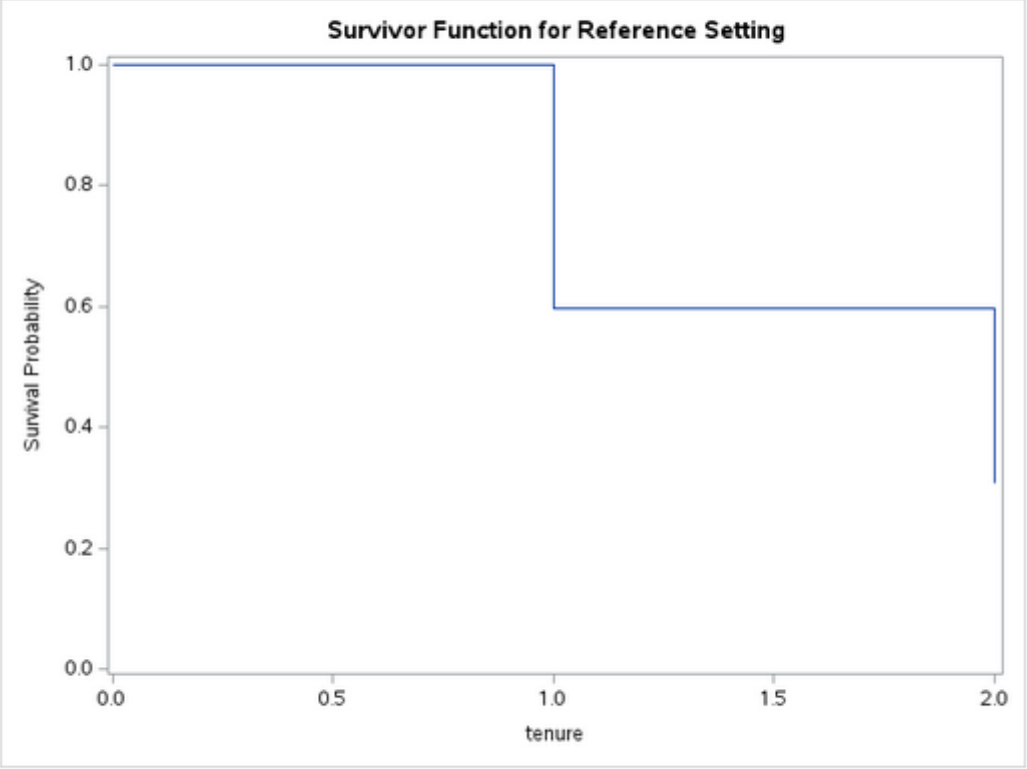


Supremum Test for Functional Form				
Variable	Maximum Absolute Value	Replications	Seed	Pr > MaxAbsVal
Risk_appetite	0.2224	1000	2025663561	0.9500
Inv_potential	0.2623	1000	2025663561	0.9350





Supremum Test for Proportionals Hazards Assumption				
Variable	Maximum Absolute Value	Replications	Seed	Pr > MaxAbsVal
Risk_appetite	0.3142	1000	2025663561	0.4930
Inv_potential	0.6769	1000	2025663561	0.2050



Reference Set of Covariates for Plotting						
Risk_appetite	FUND_PERFORMANCE	Inv_potential	Inv_involvement	AUM	Complex_prod	Complaints
2.0333333333	1.7033333333	1.73	1.74	2.2033333333	0.6333333333	0.9033333333